

IMO PHASE II LEVEL 2

DOUBLE COUNTING(With Solution)

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19th November, 2022

Double Counting: The main idea is to count some quantity in two different ways. Often, the main difficulty of a double counting problem is figuring out the specific quantity to double count. It can be used to prove some equalities (or inequalities) or to find some bounds for the solution.

Example 1. Let $p_n(k)$ be the number of permutations of the set $S = \{1, 2, 3, \dots, n\}$ which have exactly k fixed points. Prove that

$$\sum_{k=0}^n kp_n(k) = n!.$$

(A permutation p is a bijective map from $S \rightarrow S$. An element $s \in S$ is called a fixed point of p if $p(s) = s$.)

Ans. Let X be the collection of all pairs (s, p) where $p \in \text{Sym}_n$ is a permutation with $s \in S$ being one of its fixed points. We will count X in two ways:

LHS: For any $p \in \text{Sym}_n$ with exactly k fixed points, it admits k such pairs in X . Hence $|X| = \sum_{k=0}^n kp_n(k)$.

RHS: For any $s \in S$, there are obviously $(n-1)!$ permutations fix s . Hence $|X| = n \cdot (n-1)!$.

Example 2. (China TST 1992, problem 1) 16 students took part in a competition. All problems were multiple choice style. Each problem had four choices. It was said that any two students had at most one answer in common, find the maximum number of problems.

Ans. Let the students be $S_1, S_2, \dots, S_{16}$ and the questions be $Q_1, Q_2, \dots, Q_n$. Define X as the number of triples (S_i, S_j, Q_k) where $1 \leq i < j \leq 16$ such that S_i and S_j got the same answer for Q_k .

Because any two students have at most one answer in common, we have

$$X \leq 1 \cdot \binom{16}{2} = 120.$$

Now, we find a lower bound for X by analyzing each question individually.

For each Q_k , suppose a people answered "A" and so on. Since $a + b + c + d = 16$, Cauchy-Schwarz implies

$$\begin{aligned} \binom{a}{2} + \binom{b}{2} + \binom{c}{2} + \binom{d}{2} &= \frac{a^2 + b^2 + c^2 + d^2 - (a + b + c + d)}{2} \\ &\geq \frac{(a + b + c + d)^2}{8} - 8 = 24. \end{aligned}$$

Thus, there are at least 24 triples of the form (S_i, S_j, Q_k) . Now, symmetry implies

$$24n \leq X \leq 120 \implies n \leq 5.$$

Problem 1. A stage course is attended by $n \geq 4$ students. The day before the final exam, each group of three students conspire against another student to throw him/her out of the exam. Prove that there is a student against whom there are at least $\sqrt[3]{(n-1)(n-2)}$ conspirators.

Ans. We count the number of triplets $T = (A, B, C, D)$ where (A, B, C) conspire against D . First we choose (A, B, C) , this can be done in $\binom{n}{3}$ ways and by the given hypothesis we get there is at least one such D . Hence $T \geq \binom{n}{3}$. Let the number of conspirators against particular person i be at most k . Then after choose D we can select (A, B, C) in at most $\binom{k}{3}$ ways so $T < n \cdot \binom{k}{3}$. Therefore

$$n \cdot \binom{k}{3} > \binom{n}{3} \implies k^3 > k(k-1)(k-2) > (n-1)(n-2) \implies k > \sqrt[3]{(n-1)(n-2)}.$$

Hence there is a person against whom at least $\sqrt[3]{(n-1)(n-2)}$ people conspired and we are done.

Problem 2. (LMNas UGM 32 Final Round) Determine the smallest natural number n such that there exist sets $A_1, A_2, \dots, A_n \subseteq \{1, 2, 3, \dots, n\}$ satisfying the following criteria:

1. $i \notin A_i$ for every $i = 1, 2, \dots, n$;
2. $i \in A_j$ if and only if $j \notin A_i$ for every distinct i and j where $i, j \in \{1, 2, \dots, n\}$;
3. $A_i \cap A_j$ is not an empty set for every $i, j \in \{1, 2, \dots, n\}$.

Ans. The answer is $\boxed{n=7}$. First of all, let's count $\sum_{i=1}^n |A_i|$. Let's transfer each number $(1-n)$ and each set $(A_1 - A_n)$ to a bipartite graph. Connect to points if 1 of them is number, another is set and this number is in this set. Not let's see this quadruplet $\{i, j, A_i, A_j\}$. From the statement, we get this subgraph has 1 edge and since we can choose $\{i, j\}$ in $\frac{n(n-1)}{2}$ ways there are $\frac{n(n-1)}{2}$ edges in this graph. So, $\sum_{i=1}^n |A_i| = \frac{n(n-1)}{2}$. Obviously $n=1, 2$ doesn't work. Assume $n \geq 3$. Now assume that there is a set with less than 3 elements.

1. If there is a set with exactly 1 element: WLOG it be $A_1 = \{2\}$. Since $A_1 \cap A_2 \neq \emptyset$ we get $2 \in A_2$ which is contradiction.
2. If there is a set with exactly 2 element: WLOG it be $A_1 = \{2, 3\}$. Since $A_1 \cap A_2, A_3 \neq \emptyset$ we get $3 \in A_2$ and $2 \in A_3$, which is not allowed. Contradiction.

So we get $|A_i| \geq 3 \implies \sum_{i=1}^n |A_i| = \frac{n(n-1)}{2} \geq 3n \implies n \geq 7$. Construction for $n=7$ can be like this : $A_1 = \{2, 3, 4\}$ $A_2 = \{3, 5, 7\}$ $A_3 = \{4, 6, 7\}$ $A_4 = \{2, 5, 6\}$ $A_5 = \{1, 3, 6\}$ $A_6 = \{1, 2, 7\}$ $A_7 = \{1, 4, 5\}$

Problem 3. (Romanian Master Of Mathematics 2012) Given a positive integer $n \geq 3$, colour each cell of an $n \times n$ square array with one of $\lfloor (n+2)^2/3 \rfloor$ colours, each colour being used at least once. Prove that there is some 1×3 or 3×1 rectangular subarray whose three cells are coloured with three different colours.

Ans. We prove that if there is no 3-colored such subarray, then the number of distinct colors M on the board is at most $\lfloor (n+2)^2/3 \rfloor - 1$.

Given an $n \times n$ board, let a 3×1 or 1×3 tile be called a "supertile" if at least one of its cells lies on the board (so the supertiles are the set of all 3×1 or 1×3 tiles on the board, plus the ones that are hanging off of the sides).

Fix a coloring of the board and assume that there is no 3×1 or 1×3 tile colored by three different colors. We will count the set X of pairs of a supertile and a color c which appears in a cell covered by that supertile.

On the one hand – counting by supertiles – each supertile must then be incident to at most two colors. Even more: the ones which hang off the board with only one covered cell actually on the board have only one incident color. So $|X| \leq 2(n+2)^2 - 4n = 2n^2 + 4n + 8$.

On the other hand – counting by colors – for any color c appearing on the board in some cell x , the color c is incident to at least 6 different supertiles: the horizontal one with x in the leftmost position, the horizontal one with x in the center position, the horizontal one with x in the rightmost position, and all the corresponding ones for the vertical supertiles. So $6M \leq |X|$.

$$\text{Thus, } 6M \leq 2n^2 + 4n + 8 \implies M \leq (n^2 + 2n + 4)/3 \leq \lfloor (n+2)^2/3 \rfloor - 1.$$

Problem 4. Prove the identity:

$$\sum_{k=0}^n \binom{n}{k} \cdot \binom{n+k}{k} = \sum_{k=0}^n 2^k \cdot \binom{n}{k}^2$$

Ans. We claim that both sides are calculating the number of ways to choose an n -person subset from n boys and n girls, with some boys labeled special.

RHS: casework on the number of boys. Say we have n boys and n girls and want to choose an n -person subset of people. There are $\binom{n}{k}$ ways to choose k boys and $\binom{n}{n-k} = \binom{n}{k}$ ways to choose the girls. After choosing our subset, we can label some boys special; there are 2^k ways for this. Combining, we get $\sum_{k=0}^n 2^k \binom{n}{k}^2$.

LHS: casework on the number of boys that are special. There are $\binom{n}{k}$ ways to label k boys special. Then there are $\binom{2n-k}{n-k}$ ways to complete our n -person subset. Combining, we get $\sum_{k=0}^n \binom{n}{k} \binom{2n-k}{n-k}$. If we rewrite $\binom{n}{k}$ as $\binom{n}{n-k}$ and replace $n-k$ with k , we obtain the desired sum $\sum_{k=0}^n \binom{n}{k} \binom{n+k}{k}$.

Problem 5. Prove that

$$\sum_{i=0}^n \sum_{j=0}^n \frac{(n+i+j)!}{n!i!j!} 3^{-i-j} = 3^n.$$

Ans. Suppose 3 players A, B, and C are playing a game repeatedly. Each has a $1/3$ probability of winning a round. What is the probability that A is the first to win $n+1$ rounds? By symmetry, it's $1/3$.

On the other hand, we'll constructively count the cases where A is the first. Suppose that B has won i rounds and C has won j rounds when A wins their last round. This means that out of the first $n+i+j$ rounds, A won n rounds, and then won the $(n+i+j+1)$ st round. There are $\frac{(n+i+j)!}{n!i!j!}$ ways to arrange the wins of A,B,C in all rounds except the last, and the probability of a particular sequence of round wins occurring is $3^{-n-i-j-1}$.

Therefore,

$$\sum_{i=0}^n \sum_{j=0}^n \frac{(n+i+j)!}{n!i!j!} 3^{-n-i-j-1} = \frac{1}{3}$$

which simplifies to the desired identity.

Problem 6. $n > 2$ is an integer and $p(> 2n-4)$ a prime. Let M be a set of n points on a plane. No three points are collinear. Integers are written on each point, satisfying following conditions.

1. There are exactly one integer which is $0 \pmod p$;
2. If A, B, C are different three points of M , sum of integers on circumcircle of $\triangle ABC$ is $0 \pmod p$.

Prove that all points of M are on a circle.

Ans. Let n integers are $0, x_1, x_2, \dots, x_{n-1} (x_i \not\equiv 0 \pmod p)$. Let $S(C)$ be sum of numbers on a circle C . Suppose there are k_i circles that include 0 and x_i . Since every 3 points lie in a circle, each $x_j \neq x_i$ lies in exactly one of the k_i circles. Denote $S = x_1 + x_2 + \dots + x_{n-1}$. For a circle which include 0 and x_i , the sum of numbers on this circle is $0 \pmod p$. Hence $\sum_{j=1}^{k_i} S(C_j) = \sum_{j=1}^{n-1} x_j - k_i x_i = 0$, or we have

$$S \equiv (-x_i) \times k_i + 0 + x_i \equiv x_i(1 - k_i) \pmod p \dots (*)$$

On the other hand, we calculate $\sum_C S(C) \equiv 0 \pmod p$ when C includes 0, x_i, x_j . From the fact that each x_i is on exactly k_i circles, $0 \equiv \sum_C S(C) \equiv \sum_i x_i k_i$.

By summing all $(*)$ for each i , we get $(n-1)S \equiv S - \sum_i k_i x_i \equiv 0$, i.e. $S \equiv 0 \pmod p$. Hence $p | x_i(1 - k_i)$ for all i . From $p \nmid x_i$ and $k_i \leq n-2$, we get $k_i = 1$ for all i . This means all points are on a same circle.

Problem 7. Let S be a set of n persons such that:

1. any person is acquainted to exactly k other persons in S ;
2. any two persons that are acquainted have exactly l common acquaintances in S ;

3. any two persons that are not acquainted have exactly m common acquaintances in S .

Prove that

$$m(n-k) - k(k-l) + k - m = 0.$$

Ans. We double count triplets (P_1, P_2, C) , where P_1 and P_2 are two people in S and C is a common acquaintance of the two.

Count 1: Pick a person P . Then, there are k people P is acquainted to. If we choose any two of these people, they clearly form a triplet of the form we want. Since there are n people total, there are

$$n \binom{k}{2}$$

such triplets.

Count 2: We now count by cases: when P_1 and P_2 are acquainted, and when they are not acquainted. Fix P_1 . If P_2 knows P_1 , there are k choices for P_2 . Since the order in which we choose these people is irrelevant and there are n choices for P_1 , there are $\frac{nk}{2}$ such pairs P_1, P_2 . We are given that P_1 and P_2 have l common acquaintances in S , so there are $\frac{nk l}{2}$ such triplets where P_1 and P_2 are acquainted.

If P_1 and P_2 are not acquainted, after fixing P_1 , there are $n-k-1$ choices for P_2 (make sure you see why this isn't $n-k$ - we don't want P_1 to be counted in the set of such P_2). Since there are n choices for P_1 , and order is irrelevant, there are $\frac{n(n-k-1)}{2}$ such pairs P_1 and P_2 . We are given that P_1 and P_2 have m common acquaintances in S , so there are $\frac{n(n-k-1)l}{2}$ such triplets where P_1 and P_2 are not acquainted.

It follows that there are

$$\frac{nk l}{2} + \frac{n(n-k-1)l}{2}$$

such triplets.

Thus,

$$\begin{aligned} \frac{nk l}{2} + \frac{n(n-k-1)l}{2} &= n \binom{k}{2} \\ kl + (n-k-1)l &= k(k-1) \\ m(n-k) - k(k-l) + k - m &= 0. \quad \square \end{aligned}$$

Problem 8. (IMO 1989 Q3) Let n and k be positive integers and let S be a set of n points in the plane such that

1. no three points of S are collinear, and
2. for every point P of S there are at least k points of S equidistant from P .

Prove that

$$k < \frac{1}{2} + \sqrt{2 \cdot n}.$$

Ans. Define $T(n)$ as the number of isosceles triangle. We double count $T(n)$. We count them in the following two ways: by apex and by vertices.

We first count by apex. Fix a given point P . Since there are at least k points equidistant from P , there are at least $\binom{k}{2}$ ways to select the other two vertices in order to get an isosceles triangle. Since there are n choices for P ,

$$T(n) \geq n \binom{k}{2}.$$

We now count by vertices. Pick two points in S , A and B . To get an isosceles triangle ABP , P must lie on the perpendicular bisector of $\overline{AB}$. Since no three points are collinear, there are at most 2 choices for P . Since there

are $\binom{n}{2}$ ways to pick A and B ,

$$T(n) \leq 2\binom{n}{2}.$$

Hence,

$$\begin{aligned} 2\binom{n}{2} &\geq n\binom{k}{2} \\ n^2 - n &\geq \frac{1}{2}n(k^2 - k) \\ k^2 - k - (2n - 2) &\leq 0. \end{aligned}$$

Thus,

$$k \leq \frac{1 + \sqrt{1 + 8n - 8}}{2} < \frac{1}{2} + \frac{1}{2}\sqrt{8n} = \frac{1}{2} + \sqrt{2n}$$

Problem 9. (IMO 1998 Q2) In a contest, there are m candidates and n judges, where $n \geq 3$ is an odd integer. Each candidate is evaluated by each judge as either pass or fail. Suppose that each pair of judges agrees on at most k candidates. Prove that

$$\frac{k}{m} \geq \frac{n-1}{2n}.$$

Ans. We double count triplets (J_p, J_q, C) , where J_p and J_q are judges, and candidate C is agreed upon by the two judges. Let T be the number of such triplets.

We first count by pairs of judges. Pick two judges, A and B . Then, there are at most k candidates A and B agree upon. Since there are $\binom{n}{2}$ choices for (A, B) , $T \leq k\binom{n}{2}$.

Next, we count by candidates. Let $a(x)$ be the number of judges who PASS candidate x . We find that $T = \sum \left(\binom{a(x)}{2} + \binom{n-a(x)}{2} \right)$, where the sum ranges across all candidates.

The rest of the solution is just boring algebra. We include it here for completeness.

By convexity, we have (recall that n is odd)

$$\binom{a(x)}{2} + \binom{n-a(x)}{2} \geq \binom{\frac{n+1}{2}}{2} + \binom{\frac{n-1}{2}}{2} = \frac{(n-1)^2}{4}.$$

Since there are m candidates, we find that

$$T \geq \frac{m(n-1)^2}{4}.$$

Thus,

$$\begin{aligned} k\binom{n}{2} &\geq \frac{m(n-1)^2}{4} \\ \frac{1}{2}kn(n-1) &\geq \frac{m(n-1)^2}{4} \\ \frac{k}{m} &\geq \frac{n-1}{2n}. \end{aligned}$$

Problem 10. (Iran Olympiad) A school has n students. Each student can participate in any number of classes that he wants. Every class has at least two students participating in it, and if two different classes have at least two common students, then the number of students in these two classes is different. Prove that the number of classes is not greater than $(n-1)^2$.

Ans. We double count pairs (C, t) , where C is a class containing exactly t students. Define a function s ,

where $s(t)$ denotes the number of classes containing exactly t students. Also, there are $\binom{t}{2}$ pairs of students in each of these classes. It follows that there are $s(t)\binom{t}{2}$ pairs of students in classes containing exactly t students. We must have $s(t)\binom{t}{2} \leq \binom{n}{2}$, so that

$$s(t) \leq \frac{\binom{n}{2}}{\binom{t}{2}},$$

for all t .

Finally, note that the number of classes is $\sum_{t=2}^n s(t)$. Letting the number of classes be N , we get

$$N = \sum_{t=2}^n s(t) \leq \sum_{t=2}^n \frac{\binom{n}{2}}{\binom{t}{2}} = \binom{n}{2} \sum_{t=2}^n \frac{2}{t(t-1)}.$$

After some partial fraction decomposition and telescoping, the last sum becomes $2 - \frac{2}{n}$. Thus,

$$N \leq \binom{n}{2} \left(2 - \frac{2}{n}\right) \leq n(n-1) \left(\frac{n-1}{n}\right) \leq (n-1)^2.$$

Problem 11. (Russia 1999) In a class, every boy knows at least one girl. Prove that there is a group containing at least half of the students such that each boy in the group knows an odd number of girls in the group.

Ans. We double count pairs $(B, \mathcal{G})$, where B is a boy knowing an odd number of girls in some set of girls $\mathcal{G}$.

We first count by boys. Let $f(B)$ denote the number of girls a given boy B knows, g be the number of girls, and s be the total number of students. It follows that there are $\binom{f(B)}{1} + \binom{f(B)}{3} + \dots = 2^{f(B)-1}$ ways to choose an odd number of girls B knows. There are then $2^{g-f(B)}$ ways to pick the girls that B doesn't know in $\mathcal{G}$. It follows that there are $2^{g-f(B)} 2^{f(B)-1} = 2^{g-1}$ possible sets $\mathcal{G}$ for B . Since there are $s - g$ boys, it follows that there are $(s - g)2^{g-1}$ pairs.

Next, we count by sets of girls. Let $A(S)$ be the set of boys knowing an odd number of girls in a set of girls S . It follows that there are $\sum_S |A(S)|$ such pairs. Therefore,

$$\sum_S |A(S)| = (s - g)2^{g-1}.$$

We claim that there must be some set S for which $|A(S) \cup S| \geq \frac{s}{2}$. We prove this by contradiction: suppose $\forall S$ that $|A(S) \cup S| < \frac{s}{2}$. Then, since there are 2^g sets of girls, $\sum_S |A(S) \cup S| < \frac{s}{2}(2^g) = s2^{g-1}$. Since $A(S)$ is a set of boys and S is a set of girls, we find that $|A(S) \cup S| = |A(S)| + |S|$. Hence,

$$\begin{aligned} \sum_S (|A(S)| + |S|) &< s2^{g-1} \\ \sum_S |A(S)| + \sum_S |S| &< s2^{g-1} \\ (s - g)2^{g-1} + \sum_S |S| &< s2^{g-1} \\ \sum_S |S| &< g2^{g-1}. \end{aligned}$$

Note that $\sum_S |S| = \sum_{k=0}^g k \binom{g}{k} = g2^{g-1}$. Contradiction. Hence it is possible to find such group.